

Mathematical Thinking

Module 1: Introduction

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Contents

Knowledge Map

Topic	Core Question
Mathematical thinking	How do we turn a problem into something provable?
Logic	Why do relationships between propositions matter more than slogans?

0. Overview

This unit starts with the basic stance of mathematical thinking and emphasizes the relationship between definitions, propositions, and proofs.

1. Introduction to Mathematical Thinking

Review:[Introduction to Mathematical Thinking](#)(12:34)

The first lecture explains `mathematical thinking`: not speed, but structure, assumptions, and chains of reasoning.

Background Reading: [What Is Mathematical Thinking?](#) 01_intro_background_reading.pdf

Summary

- First English summary point
- Second English summary point

2. Logic and Proof

Review:[Logic and Proof](#)(18:20)

The second lecture moves into `logic` and `proof`, showing why a valid argument must make premises and conclusions explicit.

3. Key Takeaways

1. Define first, then reason.
2. Unclear premises make conclusions fragile.
3. A counterexample is stronger than a slogan.

4. Exercises

A. Multiple Choice (1 questions)

1. Which statement is a valid definition?

- A. A vague description B. A precise criterion C. A personal opinion D. A slogan

B. True/False (1 questions)

1. A proof can skip its assumptions when they are obvious.()

C. Short Answer (1 questions)

1. Explain the role of a counterexample.

5. Answer Key**Multiple Choice**

1.B

True/False

1. False

Short Answer Key Points

1. A counterexample shows that a general claim does not hold.